JAVA PROJECT REPORT ON

SCHOLARSHIP ELIGIBILITY VERIFICATION AND STUDENT SHORTLISTING SYSTEM

Submitted by

Aswin Kumar K

[25BCS207]

Under the Guidance of

[DHARANEESH.S]

[TECHNICAL TRAINER]

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DECLARATION

I hereby declare that the project entitled **"Scholarship Eligibility Verification and Student Shortlisting System"** submitted in partial fulfillment of the requirements for the course **[Java Programming]** is a record of my work carried out under the guidance of my project guide at KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Place: Coimbatore

Aswin Kumar K

CERTIFICATE

This is to certify that the project entitled “Scholarship Eligibility Verification and Student Shortlisting System” is a bona fide work carried out by Aswin Kumar K, B.Sc. Computer Science, KG College of Arts and Science, in partial fulfillment of the requirements for the course Java Programming.

This project has been completed under the supervision and guidance of the project guide during the academic year 2026–2027.

Place: Coimbatore

Date: _____

Project Guide	Head of the Department
_____	_____

TABLE OF CONTENTS

S. No.	Topics	Page No.
1	Introduction	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	Program Analysis	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	System Design	
3.1	Modules	
4	Methodology	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	Source Code Explanation	
6	Input and Output Screens	
7	Discussion and Conclusion	
7.1	Discussion	
7.2	Conclusion	
8	References	

ABSTRACT

The Scholarship Eligibility Verification and Student Shortlisting System is a Java-based application designed to automate the process of verifying student eligibility for scholarships and preparing a shortlist of qualified candidates.

In traditional scholarship selection processes, checking academic marks, family income, and other requirements manually can be time-consuming and prone to errors. Managing many applications also makes it difficult to identify eligible students efficiently.

This system allows users to enter student details such as name, register number, academic percentage, and annual family income. It evaluates these details against predefined scholarship criteria and determines eligibility.

The application can generate a shortlist of eligible students and rank them according to academic performance. The project demonstrates the practical application of object-oriented programming to improve the efficiency and reliability of scholarship selection.

1. INTRODUCTION

Scholarships provide financial assistance to students who meet academic, financial, or other eligibility requirements. Educational institutions often receive applications from many students, and checking each application manually requires considerable time and effort.

Manual verification can lead to errors in checking academic performance, family income, and eligibility conditions. Preparing a shortlist also becomes difficult when the number of applicants increases.

To overcome these challenges, the Scholarship Eligibility Verification and Student Shortlisting System is developed using Java. It automates eligibility verification and identifies students who satisfy predefined scholarship criteria.

1.1 Objective

- To design and develop a scholarship eligibility verification system.
- To automate the process of checking student eligibility.
- To verify academic percentage and annual family income.
- To reduce errors in manual verification.
- To prepare a shortlist of eligible students.
- To rank eligible students based on academic performance.
- To implement object-oriented programming concepts using Java.
- To improve the speed and accuracy of scholarship selection.

1.2 Overview

This menu-driven Java application manages basic scholarship screening operations. It accepts student information and checks whether the student satisfies the predefined eligibility requirements. Eligible students are added to a shortlist arranged by academic percentage.

- Student data entry.
- Academic percentage verification.
- Annual family income verification.
- Scholarship eligibility checking.
- Student shortlisting and ranking.
- Display of eligibility results and reports.

1.3 Features

- Easy entry of student details.
- Automatic eligibility verification.
- Checking of academic and financial criteria.
- Clear eligible and ineligible status.
- Automatic shortlist generation.
- Ranking by academic percentage.

- Organized display of student records.
- Fast processing of multiple applications.

2. PROGRAM ANALYSIS

2.1 Problem Definition

Traditional scholarship selection processes face several challenges:

- Manual verification is time-consuming.
- Checking marks and family income requires effort.
- Errors may occur during eligibility verification.
- Preparing a shortlist manually is difficult.
- Managing many student records is challenging.
- Comparing academic performance takes time.

These problems highlight the need for an automated system that verifies scholarship eligibility and prepares a shortlist efficiently.

2.2 Project Overview

The project provides a Java-based solution for scholarship eligibility verification and student shortlisting. It collects student information and compares it with predefined scholarship conditions. Students who satisfy the conditions are marked eligible; others are marked ineligible.

- Stores student details during program execution.
- Verifies eligibility based on predefined criteria.
- Displays eligibility status.
- Identifies qualified candidates.
- Ranks eligible students by academic percentage.
- Displays the shortlist and summary.

2.3 Scopes and Limitations

Scope:

- Colleges and educational institutions.
- Scholarship screening departments.
- Student financial assistance programs.
- Academic mini-project demonstrations.

Limitations:

- Console-based interface without a GUI.
- No permanent database in the basic version.
- Eligibility depends on predefined criteria.

- Document verification is not included.
- Supports only basic scholarship selection.

3. SYSTEM DESIGN

The system is divided into modules to ensure proper organization and efficient functioning.

3.1 Modules

1. Student Module

Manages student name, register number, academic percentage, annual family income, and other required details.

2. Scholarship Criteria Module

Defines minimum academic percentage, maximum annual family income, and any additional conditions required by the scholarship scheme.

3. Eligibility Verification Module

Compares student information with eligibility requirements and determines the result.

4. Student Shortlisting Module

Identifies eligible students and prepares a shortlist ranked by academic percentage.

5. Report Module

Displays student details, eligibility status, shortlist, academic ranking, total applicants, and eligible/ineligible counts.

4. METHODOLOGY

The system follows a structured methodology to ensure accuracy, reliability, and ease of use.

4.1 Problem Analysis

The first step identifies problems in manual scholarship verification, including time-consuming checks, difficulty managing student information, and errors in preparing a shortlist. The proposed system addresses these problems through automation.

4.2 System Design

The system is divided into modules for student data entry, scholarship criteria, eligibility verification, shortlisting, and report generation. Each module performs a specific task.

4.3 Program Development

The system is implemented using Java and core programming concepts:

- Classes and objects.
- Constructors and methods.
- Conditional statements.
- Loops.
- ArrayList for student records during execution.
- Sorting techniques for ranking eligible students.
- Modular programming.

4.4 Testing and Validation

The system is tested using different student details to verify that the eligibility conditions work correctly.

- Students who satisfy all eligibility conditions.
- Students with marks below the minimum requirement.
- Students whose family income exceeds the limit.
- Students who fail multiple criteria.
- Eligible students with different academic percentages.
- Invalid input values.

Testing helps ensure correct eligibility results and an accurate shortlist.

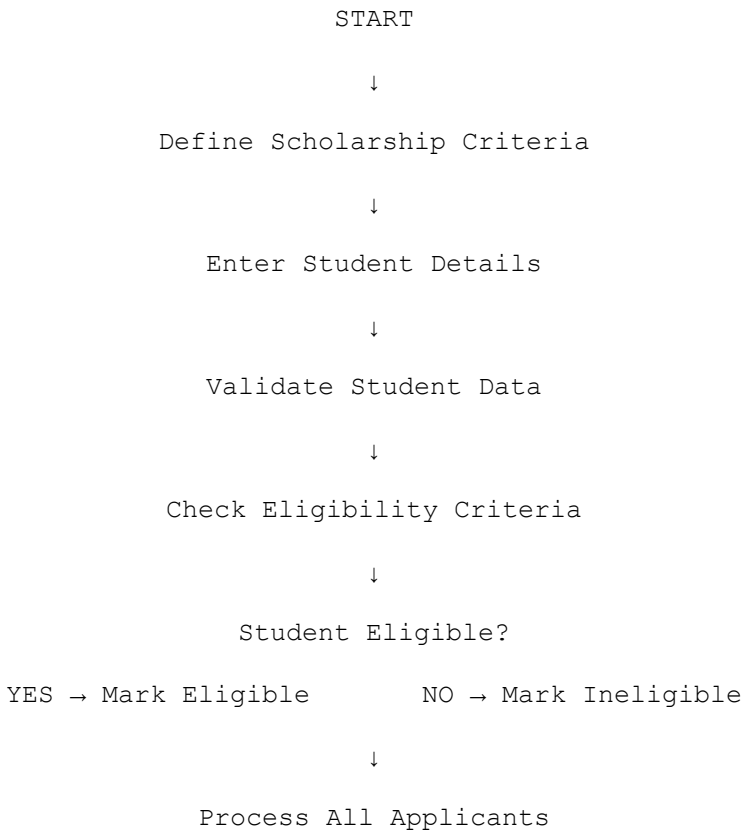
4.5 Result and Documentation

The final stage involves executing the program, verifying its output, and documenting its functionality. The system displays each student's eligibility and generates a shortlist based on the defined criteria.

4.6 Algorithm

1. Start the program.
2. Define scholarship eligibility criteria.
3. Enter the number of applicants.
4. Enter student details.
5. Validate the entered information.
6. Check academic percentage.
7. Check annual family income.
8. Determine whether the student satisfies the criteria.
9. Mark the student eligible or ineligible.
10. Store student details and result.
11. Repeat for all applicants.
12. Prepare a shortlist of eligible students.
13. Sort eligible students by academic percentage.
14. Display eligibility results and shortlist.
15. End the program.

4.7 Flowchart (Text Representation)



↓

Prepare Eligible Shortlist

↓

Rank by Academic Marks

↓

Display Final Results

↓

STOP

5. SOURCE CODE EXPLANATION

The system uses a modular, object-oriented approach. Classes manage student information, eligibility checking, and the overall application workflow.

1. Student Class

Stores and manages student-related information.

Attributes: Name, register number, academic percentage, annual family income, and eligibility status.

Functions: Accepts student details, stores information, provides methods to access data, and displays student details and eligibility status.

Importance: Organizes the information required for scholarship eligibility verification.

2. Scholarship Class

Manages scholarship eligibility criteria.

Attributes: Minimum academic percentage and maximum annual family income.

Functions: Defines conditions, compares student details with requirements, and determines eligibility.

Importance: Ensures eligibility decisions follow the predefined scholarship rules.

3. Shortlisting Class

Identifies qualified students and organizes them into a shortlist.

- Selects eligible students.
- Collects eligible student records.
- Sorts students by academic percentage.
- Displays the ranked shortlist.

Importance: Simplifies selection and helps identify students who meet scholarship requirements.

4. Main Class

Controls application execution.

- Displays menu and instructions.
- Accepts student details.
- Calls eligibility verification methods.
- Manages student records.
- Generates the shortlist.
- Displays final results.

Typical workflow:

16. The user enters student details.
17. The system checks eligibility.
18. The student is marked eligible or ineligible.
19. The process continues for all applicants.
20. Eligible students are shortlisted and ranked.
21. The final results are displayed.

Concepts used: Object-oriented programming, conditional statements, loops, ArrayList, sorting, methods, and modular programming.

6. INPUT AND OUTPUT SCREENS

Input

The system accepts the following information:

- Student name.
- Register number.
- Academic percentage.
- Annual family income.
- Number of applicants.

Minimum academic percentage and maximum family income are defined according to the scholarship scheme.

Output

The system displays:

- Student details.
- Eligibility status.
- Reason for ineligibility, where applicable.
- Shortlisted student list.
- Ranking by academic percentage.
- Total applicants.
- Number of eligible students.
- Number of ineligible students.

Sample Output

```
SCHOLARSHIP ELIGIBILITY VERIFICATION SYSTEM

Student Name         : Anitha
Register Number      : CS101
Academic Percentage  : 82%
Annual Family Income: Rs. 1,80,000

Eligibility Status   : ELIGIBLE

-------------------------------------
STUDENT SHORTLIST
-------------------------------------
Rank  Name        Percentage
1     Anitha      82%

Total Applicants    : 1
Eligible Students   : 1
Ineligible Students: 0
```

Note: This output is an example only. Actual eligibility depends on the criteria defined for the scholarship scheme.

7. DISCUSSION AND CONCLUSION

7. RESULTS AND DISCUSSION

The system demonstrates how Java can automate scholarship screening. It checks student details against predefined criteria, identifies qualified candidates, and generates a shortlist based on academic performance. Automation can reduce manual effort and organize applicant information efficiently.

7.1 Discussion

This project demonstrates the practical application of Java programming to an educational administration problem. It simplifies eligibility checking, reduces repetitive work, and prepares a structured shortlist. Its modular design makes the program easier to understand and maintain.

The system's decisions depend on the accuracy of the entered student details and the criteria configured for the scholarship.

7.2 Conclusion

The Scholarship Eligibility Verification and Student Shortlisting System is a Java-based application designed to simplify scholarship screening. It accepts student details, verifies eligibility based on academic and financial criteria, and generates a shortlist of eligible students. The project demonstrates object-oriented programming, conditional statements, loops, and sorting techniques.

The system meets its basic objectives by making scholarship verification more organized, efficient, and consistent.

Future Enhancements

- Graphical user interface using Java Swing or JavaFX.
- Database integration using MySQL.
- Online scholarship application submission.
- Document upload and verification.
- Administrator login and access control.
- Support for multiple scholarship schemes.
- PDF report generation.

- Email or SMS notifications for applicants.

8. REFERENCES

The following resources can be consulted during the development of this project.

Books

- Herbert Schildt — Java: The Complete Reference.

Websites

- Oracle Java Documentation: https://docs.oracle.com/en/java/
- GeeksforGeeks: https://www.geeksforgeeks.org/java/
- W3Schools Java Tutorial: https://www.w3schools.com/java/

- JavaTpoint: https://www.javatpoint.com/java-tutorial

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